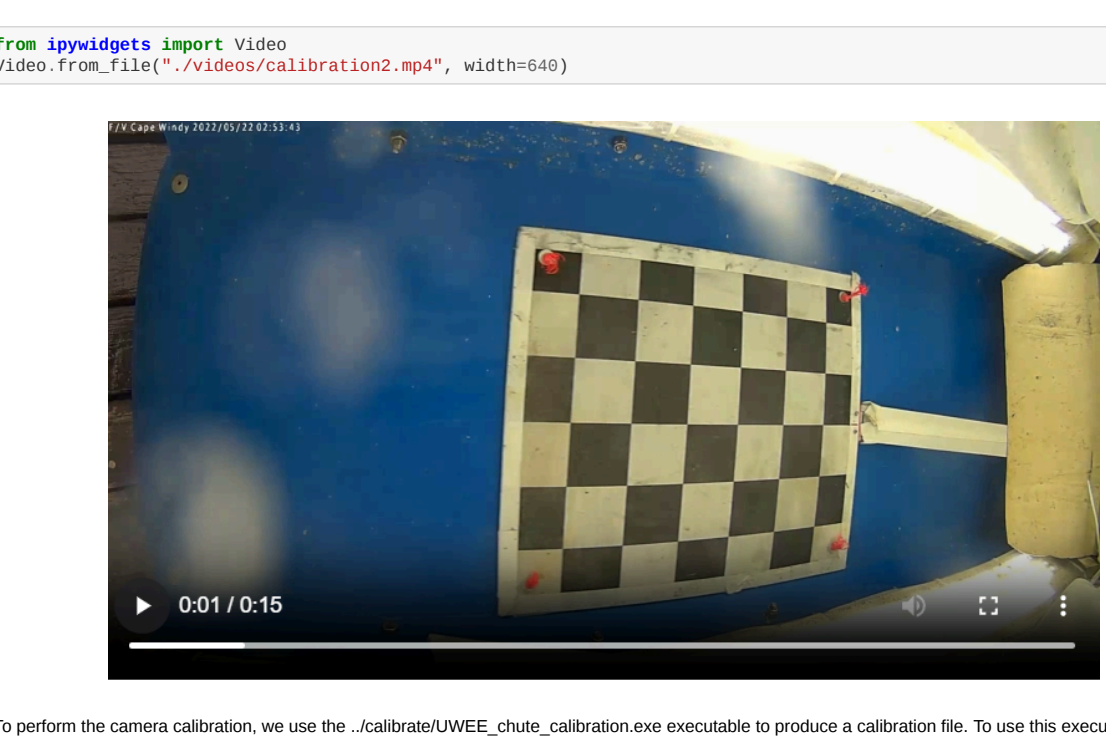
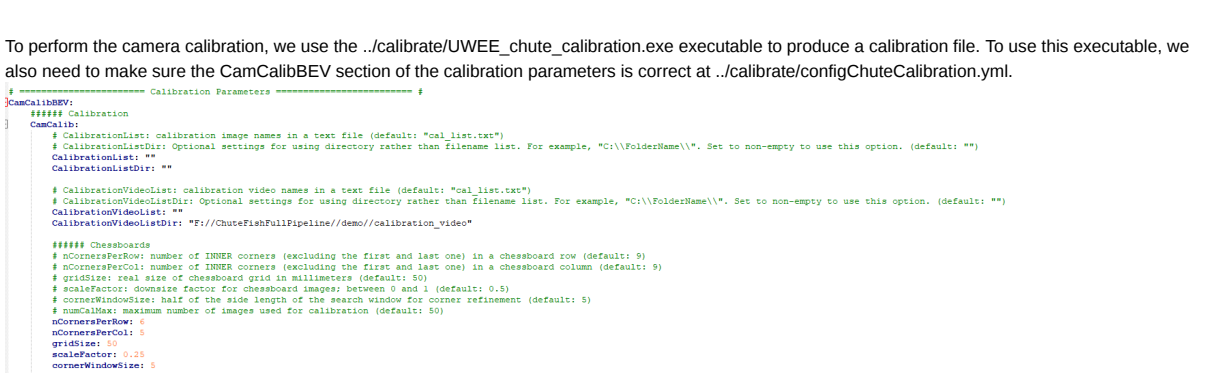
Step 1 - Camera Calibration

First, we need a calibration clip to get camera parameters. The calibration clip should have a checker board pattern that moves around on the chute floor.

In [1]:
```python
from ipywidgets import Video
Video.from_file("./videos/calibration2.mp4", width=640)
```



To perform the camera calibration, we use the ../calibrate/UWEE_chute_calibration.exe executable to produce a calibration file. To use this executable, we also need to make sure the CamCalibBEV section of the calibration parameters is correct at ../calibrate/configChuteCalibration.yml.



```
# ====================== Calibration Parameters ========================= #
CamCalibBEV:
    ###### Calibration
    CamCalib:
        # CalibrationList: calibration image names in a text file (default: "cal_list.txt")
        # CalibrationListDir: Optional settings for using directory rather than filename list. For example, "C:\\FolderName\\". Set to non-empty to use this option. (default: "")
        CalibrationList: ""
        CalibrationListDir: ""

        # CalibrationVideoList: calibration video names in a text file (default: "cal_list.txt")
        # CalibrationVideoListDir: Optional settings for using directory rather than filename list. For example, "C:\\FolderName\\". Set to non-empty to use this option. (default: "")
        CalibrationVideoList: ""
        CalibrationVideoListDir: "F://ChuteFishFullPipeline//demo//calibration_video"

        ###### Chessboards
        # nCornersPerRow: number of INNER corners (excluding the first and last one) in a chessboard row (default: 9)
        # nCornersPerCol: number of INNER corners (excluding the first and last one) in a chessboard column (default: 9)
        # gridSize: real size of chessboard grid in millimeters (default: 50)
        # scaleFactor: downsize factor for chessboard images; between 0 and 1 (default: 0.5)
        # cornerWindowSize: half of the side length of the search window for corner refinement (default: 5)
        # numCalMax: maximum number of images used for calibration (default: 50)
        nCornersPerRow: 6
        nCornersPerCol: 5
        gridSize: 50
        scaleFactor: 0.25
        cornerWindowSize: 5
        numCalMax: 16

    ###### BEV Calibration
    # CalibrationBEVList: BEV calibration image names to define the x- and y- direction
    # CalibrationListDirBEV: Optional settings for using directory rather than filename list. For example, "C:\\FolderName\\". Set to non-empty to use this option. (default: "")
    CalibrationListBEV: ""
    CalibrationListDirBEV: ""

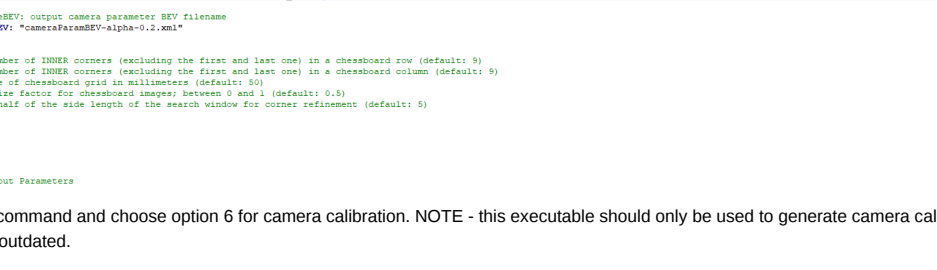
    # CalibrationVideoBEVList: BEV calibration video names to define the x- and y- direction
    # CalibrationVideoListDirBEV: Optional settings for using directory rather than filename list. For example, "C:\\FolderName\\". Set to non-empty to use this option. (default: "")
    CalibrationVideoListBEV: ""
    CalibrationVideoListDirBEV: "F://ChuteFishFullPipeline//demo//calibration_video"

    # CameraParamFilenameBEV: output camera parameter BEV filename
    CameraParamFilenameBEV: "cameraParamBEV-alpha-0.2.xml"

    ###### Chessboards
    # nCornersPerRow: number of INNER corners (excluding the first and last one) in a chessboard row (default: 9)
    # nCornersPerCol: number of INNER corners (excluding the first and last one) in a chessboard column (default: 9)
    # gridSize: real size of chessboard grid in millimeters (default: 50)
    # scaleFactor: downsize factor for chessboard images; between 0 and 1 (default: 0.5)
    # cornerWindowSize: half of the side length of the search window for corner refinement (default: 5)
    nCornersPerRow: 6
    nCornersPerCol: 5
    gridSize: 50
    scaleFactor: 0.25
    cornerWindowSize: 5

    ###### BEV Image Layout Parameters
```

Run the terminal command and choose option 6 for camera calibration. NOTE - this executable should only be used to generate camera calibration files. The other options are outdated.



```
UWEE ChuteLogger Fish Segmentation
  1) segmentation and length measurement
  2) camera calibration
  3) test background
  4) label segmentation ground truth
  ------------------------------------------
  ### Video input
  5) segmetation and length measurement
  6) camera calibration
  ------------------------------------------
  7) quit
Select one mode to continue: 6
Reading chessboard videos
```

In [17]:
```
%%cmd
"../calibrate/UWEE_chute_calibration.exe" ../calibrate/configChuteCalibration.yml
6
```

```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

(yolov8) F:\ChuteFishFullPipeline\demo>"../calibrate/UWEE_chute_calibration.exe" ../calibrate/configChuteCalibration.
yml
Using configuration file: ../calibrate/configChuteCalibration.yml
CamCalibBEV > CamCalib > CalibrationList:
CamCalibBEV > CamCalib > CalibrationListDir:
CamCalibBEV > CamCalib > CalibrationVideoList:
CamCalibBEV > CamCalib > CalibrationVideoListDir: F://ChuteFishFullPipeline//demo//calibration_video
CamCalibBEV > CamCalib > [ - ERROR - ] Cannot find the node: CameraParamFilename
CamCalibBEV > CamCalib > nCornersPerRow: 6
CamCalibBEV > CamCalib > nCornersPerCol: 5
CamCalibBEV > CamCalib > gridSize: 50
CamCalibBEV > CamCalib > scaleFactor: 0.25
CamCalibBEV > CamCalib > cornerWindowSize: 5
CamCalibBEV > CamCalib > numCalMax: 16
CamCalibBEV > CalibrationListBEV:
CamCalibBEV > CalibrationListDirBEV:
CamCalibBEV > CalibrationVideoListBEV:
CamCalibBEV > CalibrationVideoListDirBEV: F://ChuteFishFullPipeline//demo//calibration_video
CamCalibBEV > CameraParamFilenameBEV: cameraParamBEV-alpha-0.2.xml
CamCalibBEV > nCornersPerRow: 6
CamCalibBEV > nCornersPerCol: 5
CamCalibBEV > gridSize: 50
CamCalibBEV > scaleFactor: 0.25
CamCalibBEV > cornerWindowSize: 5
CamCalibBEV > [ - ERROR - ] Cannot find the node: numCalMax
CamCalibBEV > mBEV: 1
CamCalibBEV > alphaBEV: 0.2
CamCalibBEV > widthBEV: 1500
CamCalibBEV > heightBEV: 700
CamCalibBEV > xBEV: 450
CamCalibBEV > yBEV: 200
ObjSeg > GMM > method: 0
ObjSeg > GMM > maxNumBgImgs: 100
ObjSeg > GMM > isColorTransfer: 0
ObjSeg > GMM > colorTransferType: 0
ObjSeg > GMM > colorTransferCountsMax: 100
ObjSeg > GMM > VarThreshold: 16
ObjSeg > GMM > NMixtures: 5
ObjSeg > GMM > BackgroundRatio: 0.9
ObjSeg > GMM > VarThresholdGen: 9
ObjSeg > GMM > VarInit: 15
ObjSeg > GMM > VarMin: 4
ObjSeg > GMM > VarMax: 75
ObjSeg > GMM > fTau: 0.5
ObjSeg > GMM > threshHSV: 15
ObjSeg > GMM > aH: 1
ObjSeg > GMM > aS: 0.1
ObjSeg > GMM > aV: 0.2
ObjSeg > GMM > h_bg_init: 120
ObjSeg > GMM > h_bg_diff_thresh_init: 20
ObjSeg > GMM > v_bg_min_init: 100
ObjSeg > GMM > v_bg_max_init: 200
ObjSeg > GMM > nH: 1
ObjSeg > GMM > nS: 1.5
ObjSeg > GMM > nV: 1.5
ObjSeg > HistBackprojection > isUsingHSVHist: 1
ObjSeg > HistBackprojection > nbinsB: 16
ObjSeg > HistBackprojection > nbinsG: 16
ObjSeg > HistBackprojection > nbinsR: 16
ObjSeg > HistBackprojection > nbinsH: 18
ObjSeg > HistBackprojection > nbinsS: 16
ObjSeg > HistBackprojection > nbinsV: 16
ObjSeg > HistBackprojection > theta: 0.8
ObjSeg > HistBackprojection > dilationSize: 9
ObjSeg > ActiveContour > numPoints: 200
ObjSeg > ActiveContour > maxIterInner: 300
ObjSeg > ActiveContour > maxIterOuter: 3
ObjSeg > ActiveContour > wEdge: 0.6
ObjSeg > ActiveContour > wCont: 0.15
ObjSeg > ActiveContour > wCurv: 0.2
ObjSeg > ActiveContour > wBall: 0.15
ObjSeg > ActiveContour > stepSize: 9
ObjSeg > ActiveContour > minDist: 5
ObjSeg > ObjSegCRF > CRFinHSV: 1
ObjSeg > ObjSegCRF > nIterations: 5
ObjSeg > ObjSegCRF > gt_prob: 0.52
ObjSeg > ObjSegCRF > sx: 5
ObjSeg > ObjSegCRF > sy: 5
ObjSeg > ObjSegCRF > wxy: 5
ObjSeg > ObjSegCRF > sx_bgr: 60
ObjSeg > ObjSegCRF > sy_bgr: 60
ObjSeg > ObjSegCRF > sb_bgr: 40
ObjSeg > ObjSegCRF > sg_bgr: 20
ObjSeg > ObjSegCRF > sr_bgr: 20
ObjSeg > ObjSegCRF > wbgr: 20
ObjSeg > ObjSegCRF > sx_hsv: 30
ObjSeg > ObjSegCRF > sy_hsv: 30
ObjSeg > ObjSegCRF > sh_hsv: 5
ObjSeg > ObjSegCRF > ss_hsv: 5
ObjSeg > ObjSegCRF > sv_hsv: 10
ObjSeg > ObjSegCRF > whsv: 20
ObjSeg > isUsingHistBP: 0
ObjSeg > isUsingActiveContour: 0
ObjSeg > isUsingDenseCRF: 0
ObjSeg > morphSESize: 5
ObjSeg > minArea: 20000
ObjSeg > maxArea: -1
ObjSeg > minAspectRatio: 1
ObjSeg > maxAspectRatio: 7
Measurement > IsMidlineLength: 1
Measurement > MidlinePrecision: 10
Measurement > MidlineType: 1
Measurement > IsBlurMeasure: 0
Measurement > BlurMeasureThresh: 1000
Measurement > IsEdgeMeasure: 1
Measurement > EdgeMeasureThresh: 0.02
Measurement > IsShowBoundingBox: 1
Measurement > IsShowContour: 1
Measurement > IsShowMidline: 1
Measurement > IsShowBlurSections: 1
Measurement > IsShowEdgeSections: 1
InputAndOutput > InputList: img_list-sample.txt
InputAndOutput > BackgroundList:
InputAndOutput > GroundTruthList:
InputAndOutput > InputListDir:
InputAndOutput > BackgroundListDir: D:\noaa\data\chute\ChuteLabTesting\Variables\Background\
InputAndOutput > GroundTruthListDir:
InputAndOutput > InputVideoList:
InputAndOutput > InputVideoListDir:
InputAndOutput > CameraParamFilenameBEV: cameraParamBEV-alpha-0.2.xml
InputAndOutput > ROI:
InputAndOutput > DetectROI: 1
InputAndOutput > DetectROIType: 2
InputAndOutput > IsUsingShapeModel: 1
InputAndOutput > ShapeModelPath: .\fish shape\train_data\
InputAndOutput > NoseDirection: 0
InputAndOutput > FixRotation: 1
InputAndOutput > NumInitBgFrames: 100
InputAndOutput > LearningRate: 0.005
InputAndOutput > LearningRateFg: 0.001
InputAndOutput > DownSampleFrames: 10
InputAndOutput > FLower: 0.03125
InputAndOutput > IOUThreshold: 0.2
InputAndOutput > ROIshape: ROIshape.png
InputAndOutput > ShapeMovementThreshold: 100
InputAndOutput > StartImage: 0
InputAndOutput > StopImage: -1
InputAndOutput > CompareGT: 0
InputAndOutput > InputDir: F://ChuteFishFullPipeline//demo//
InputAndOutput > OutputDir: F://ChuteFishFullPipeline//demo//
InputAndOutput > OutputFilename: length.csv
InputAndOutput > WriteResults: 1
InputAndOutput > WriteResultsMask: 1
InputAndOutput > WriteResultsDemo: 0
InputAndOutput > ShowResults: 1
InputAndOutput > ShowResultsMask: 1
InputAndOutput > ShowResultsDemo: 0
InputAndOutput > WriteResultsRotatedSeg: 0
InputAndOutput > WriteResultsRotatedSegMask: 0
InputAndOutput > RotatedSegMarginRatio: 0.1
InputAndOutput > nLevel: 2

UWEE ChuteLogger Fish Segmentation
  1) segmentation and length measurement
  2) camera calibration
  3) test background
  4) label segmentation ground truth
  ------------------------------------------
  ### Video input
  5) segmetation and length measurement
  6) camera calibration
  ------------------------------------------
  7) quit
Select one mode to continue: Reading chessboard videos ...
F://ChuteFishFullPipeline//demo//calibration_video\calibration2.mp4
fps = 20
length = 300
width = 1920
height = 1080
0/300
1/300
2/300
3/300
4/300
5/300
6/300
7/300
8/300
9/300
10/300
11/300
12/300
13/300
14/300
15/300
16/300
17/300
18/300
19/300
20/300
21/300
22/300
23/300
24/300
25/300
26/300
27/300
28/300
29/300
30/300
31/300
32/300
33/300
34/300
35/300
36/300
37/300
38/300
39/300
40/300
41/300
42/300
43/300
44/300
45/300
46/300
47/300
48/300
49/300
50/300
51/300
52/300
53/300
54/300
55/300
56/300
57/300
58/300
59/300
60/300
61/300
62/300
63/300
64/300
65/300
66/300
67/300
68/300
69/300
70/300
71/300
72/300
73/300
74/300
75/300
76/300
77/300
78/300
79/300
80/300
81/300
82/300
83/300
84/300
85/300
86/300
87/300
88/300
89/300
90/300
91/300
92/300
93/300
94/300
95/300
96/300
97/300
98/300
99/300
100/300
101/300
102/300
103/300
104/300
105/300
106/300
107/300
108/300
109/300
110/300
111/300
112/300
113/300
114/300
115/300
116/300
117/300
118/300
119/300
120/300
121/300
122/300
123/300
124/300
125/300
126/300
127/300
128/300
129/300
130/300
131/300
132/300
133/300
134/300
135/300
136/300
137/300
138/300
139/300
140/300
141/300
142/300
143/300
144/300
145/300
146/300
147/300
148/300
149/300
150/300
151/300
152/300
153/300
154/300
155/300
156/300
157/300
158/300
159/300
160/300
161/300
162/300
163/300
164/300
165/300
166/300
167/300
168/300
169/300
170/300
171/300
172/300
173/300
174/300
175/300
176/300
177/300
178/300
179/300
180/300
181/300
182/300
183/300
184/300
185/300
186/300
187/300
188/300
189/300
190/300
191/300
192/300
193/300
194/300
195/300
196/300
197/300
198/300
199/300
200/300
201/300
202/300
203/300
204/300
205/300
206/300
207/300
208/300
209/300
210/300
211/300
212/300
213/300
214/300
215/300
216/300
217/300
218/300
219/300
220/300
221/300
222/300
223/300
224/300
225/300
226/300
227/300
228/300
229/300
230/300
231/300
232/300
233/300
234/300
235/300
236/300
237/300
238/300
239/300
240/300
241/300
242/300
243/300
244/300
245/300
246/300
247/300
248/300
249/300
250/300
251/300
252/300
253/300
254/300
255/300
256/300
257/300
258/300
259/300
260/300
261/300
262/300
263/300
264/300
265/300
266/300
267/300
268/300
269/300
270/300
271/300
272/300
273/300
274/300
275/300
276/300
277/300
278/300
279/300
280/300
281/300
282/300
283/300
284/300
285/300
286/300
287/300
288/300
289/300
290/300
291/300
292/300
293/300
294/300
295/300
296/300
297/300
298/300
299/300
300/300
calibrateCamera Error = 2.01785
Reading chessboard videos ...
F://ChuteFishFullPipeline//demo//calibration_video\calibration2.mp4
fps = 20
length = 300
width = 1920
height = 1080
0/300
1/300
2/300
3/300
4/300
5/300
6/300
7/300
8/300
9/300
10/300
11/300
12/300
13/300
14/300
15/300
16/300
17/300
18/300
19/300
20/300
21/300
22/300
23/300
24/300
25/300
26/300
27/300
28/300
29/300
30/300
31/300
32/300
33/300
34/300
35/300
36/300
37/300
38/300
39/300
40/300
41/300
42/300
43/300
44/300
45/300
46/300
47/300
48/300
49/300
50/300
51/300
52/300
53/300
54/300
55/300
56/300
57/300
58/300
59/300
60/300
61/300
62/300
63/300
64/300
65/300
66/300
67/300
68/300
69/300
70/300
71/300
72/300
73/300
74/300
75/300
76/300
77/300
78/300
79/300
80/300
81/300
82/300
83/300
84/300
85/300
86/300
87/300
88/300
89/300
90/300
91/300
92/300
93/300
94/300
95/300
96/300
97/300
98/300
99/300
100/300
101/300
102/300
103/300
104/300
105/300
106/300
107/300
108/300
109/300
110/300
111/300
112/300
113/300
114/300
115/300
116/300
117/300
118/300
119/300
120/300
121/300
122/300
123/300
124/300
125/300
126/300
127/300
128/300
129/300
130/300
131/300
132/300
133/300
134/300
135/300
136/300
137/300
138/300
139/300
140/300
141/300
142/300
143/300
144/300
145/300
146/300
147/300
148/300
149/300
150/300
151/300
152/300
153/300
154/300
155/300
156/300
157/300
158/300
159/300
160/300
161/300
162/300
163/300
164/300
165/300
166/300
167/300
168/300
169/300
170/300
171/300
172/300
173/300
174/300
175/300
176/300
177/300
178/300
179/300
180/300
181/300
182/300
183/300
184/300
185/300
186/300
187/300
188/300
189/300
190/300
191/300
192/300
193/300
194/300
195/300
196/300
197/300
198/300
199/300
200/300
201/300
202/300
203/300
204/300
205/300
206/300
207/300
208/300
209/300
210/300
211/300
212/300
213/300
214/300
215/300
216/300
217/300
218/300
219/300
220/300
221/300
222/300
223/300
224/300
225/300
226/300
227/300
228/300
229/300
230/300
231/300
232/300
233/300
234/300
235/300
236/300
237/300
238/300
239/300
240/300
241/300
242/300
243/300
244/300
245/300
246/300
247/300
248/300
249/300
250/300
251/300
252/300
253/300
254/300
255/300
256/300
257/300
258/300
259/300
260/300
261/300
262/300
263/300
264/300
265/300
266/300
267/300
268/300
269/300
270/300
271/300
272/300
273/300
274/300
275/300
276/300
277/300
278/300
279/300
280/300
281/300
282/300
283/300
284/300
285/300
286/300
287/300
288/300
289/300
290/300
291/300
292/300
293/300
294/300
295/300
296/300
297/300
298/300
299/300
300/300
(60:277/300),err = 4.22341
(60:279/300),err = 4.21203
(60:278/300),err = 4.1657
(60:277/300),err = 4.22241
(60:278/300),err = 4.23774
(60:277/300),err = 4.15896
(60:272/300),err = 4.19807
(60:279/300),err = 4.26962
(60:277/300),err = 4.16188
(60:269/300),err = 4.13305
(60:278/300),err = 4.1657
(60:280/300),err = 4.28995
(60:279/300),err = 4.22394
(60:277/300),err = 4.13867
(60:279/300),err = 4.26962
(60:279/300),err = 4.21203
(60:280/300),err = 4.21217
(60:278/300),err = 4.14306
(60:275/300),err = 4.04547
(60:277/300),err = 4.17626
(60:276/300),err = 4.06816
(60:279/300),err = 4.20297
(60:278/300),err = 4.14693
(60:278/300),err = 4.22346
(60:275/300),err = 4.16696
(60:279/300),err = 4.24948
(60:278/300),err = 4.20654
(60:277/300),err = 4.11335
(60:279/300),err = 4.17096
(60:278/300),err = 4.1657
(60:278/300),err = 4.13893
(60:280/300),err = 4.24031
(60:279/300),err = 4.20297
(60:277/300),err = 4.17714
(60:279/300),err = 4.26962
(60:272/300),err = 4.16058
(60:279/300),err = 4.27242
(60:278/300),err = 4.15193
(60:276/300),err = 4.06816
(60:280/300),err = 4.32306
(60:281/300),err = 4.2658
(60:280/300),err = 4.23773
(60:281/300),err = 4.28969
(60:277/300),err = 4.11872
(60:275/300),err = 4.21832
(60:278/300),err = 4.14693
(60:279/300),err = 4.26518
(60:278/300),err = 4.14013
(60:275/300),err = 4.04433
(60:276/300),err = 4.09914
(60:277/300),err = 4.13228
(60:279/300),err = 4.20336
(60:279/300),err = 4.20146
(60:276/300),err = 4.10055
(60:278/300),err = 4.23343
(60:279/300),err = 4.29592
(60:279/300),err = 4.244
(60:278/300),err = 4.14016
(60:279/300),err = 4.33139
(60:279/300),err = 4.26962
(60:277/300),err = 4.17626
(60:278/300),err = 4.16851
(60:280/300),err = 4.26196
(60:280/300),err = 4.23931
(60:272/300),err = 3.97351
(60:278/300),err = 4.15579
(60:277/300),err = 4.09884
(60:273/300),err = 3.98745
(60:279/300),err = 4.20146
(60:279/300),err = 4.34896
(60:279/300),err = 4.20297
(60:276/300),err = 4.06816
(60:279/300),err = 4.26962
(60:278/300),err = 4.26711
(60:279/300),err = 4.20336
(60:278/300),err = 4.15193
(60:280/300),err = 4.30937
(60:278/300),err = 4.14761
(60:278/300),err = 4.14693
(60:277/300),err = 4.13442
(60:280/300),err = 4.24561
(60:280/300),err = 4.23931
(60:280/300),err = 4.30937
(60:277/300),err = 4.13099
(60:279/300),err = 4.17096
(60:280/300),err = 4.30937
(60:277/300),err = 4.17445
(60:277/300),err = 4.11849
(60:279/300),err = 4.20336
(60:276/300),err = 4.17393
(60:277/300),err = 4.16188
(60:279/300),err = 4.1817
(60:280/300),err = 4.23931
(60:279/300),err = 4.20297
(60:279/300),err = 4.19257
(60:279/300),err = 4.22394
(60:278/300),err = 4.16455
(60:281/300),err = 4.27405
(60:278/300),err = 4.23462
(60:279/300),err = 4.20894
bestErr = 3.97351
bestErrMax = 10.0885
(err_all, err_all_max) = (5.36109, 17.1545)
(err_proj_all, err_proj_all_max) = (8.0009, 29.7821)
Saving camera parameters ...
Camera parameters is saved to F://ChuteFishFullPipeline//demo//\calibration\cameraParamBEV-alpha-0.2.xml

(yolov8) F:\ChuteFishFullPipeline\demo>
~~~

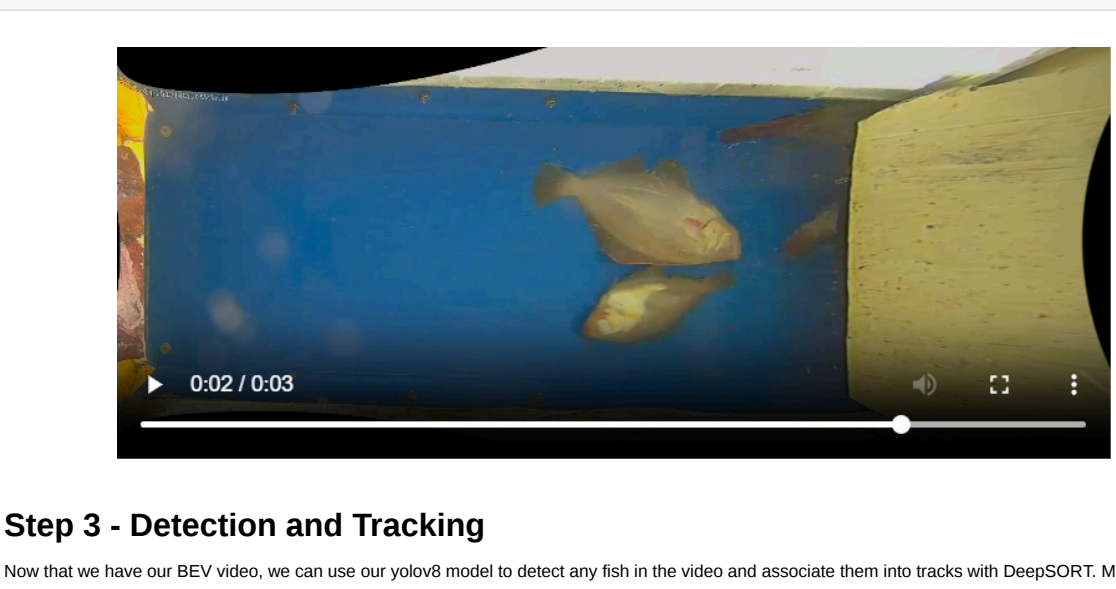
The calibration file should be saved to F:\ChuteFishFullPipeline\demo\calibration\cameraParamBEV-alpha-0.2.xml.

## Step 2 - Undistort Video

Convert video to bird's eye view (BEV) so that pixel length corresponds to real length in meters. Here is our test video.

In [2]:
```python
Video.from_file("./test_video/test2.mp4", width=640)
~~~

Set the configuration file so that our program knows where to find the input video, where to save the output results, and where the calibration file is.



```
# input video directory
input_dir: "F:\ChuteFishFullPipeline\demo\calibration\test_video"
# output directory
output_dir: "F:\ChuteFishFullPipeline\demo\calibration\test_out"
# path to camera calibration file
calib_path: "F:\ChuteFishFullPipeline\demo\calibration\cameraParamBEV-alpha-0.2.xml"

det_model_path: "yolov8_chute.pt"

save_vid: True
fps: 15

det_thresh: 0.2

class_path: chute2022.pth

segment_model: "h"
# Whether to perform segmentation or just visualize detection and classification results
segment: True
# number of points for midline
midline_pts: 10
```

In [19]:
```
%%cmd
python ../calibrate/undistort_videos.py --config_path demoChuteConfig.yml
```

```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

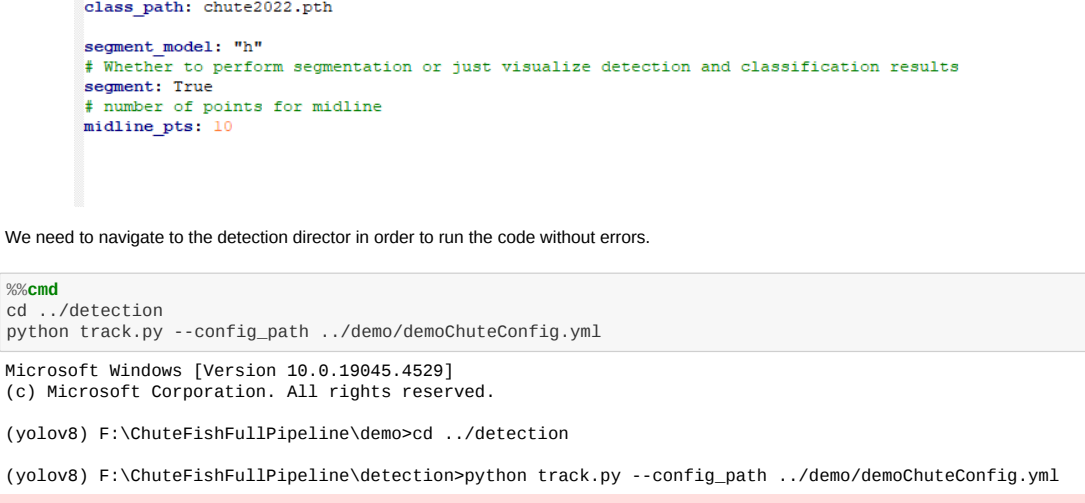
(yolov8) F:\ChuteFishFullPipeline\demo>python ../calibrate/undistort_videos.py --config_path demoChuteConfig.yml
Undistortion code running now
Processing video test2.mp4
```

```
OpenCV: FFMPEG: tag 0x30395056/'VP90' is not supported with codec id 167 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x39307076/'vp09'
```

```
Undistorting video 1/1: Frame 30/32
(yolov8) F:\ChuteFishFullPipeline\demo>
```

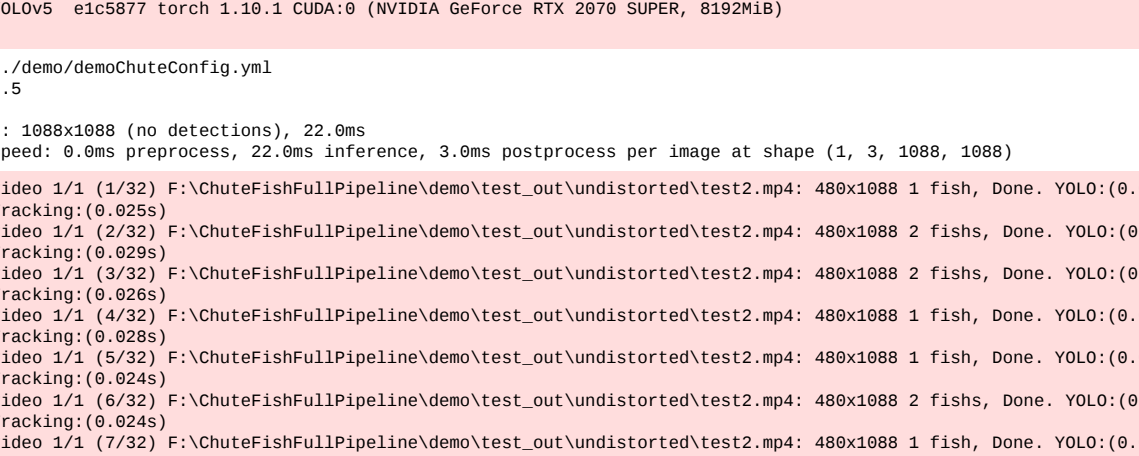
Now the BEV video should be at F:\ChuteFishFullPipeline\demo\test_out\undistorted

In [3]:
```python
from ipywidgets import Video
Video.from_file("./test_out/undistorted/test2.mp4", width=640)
```



Step 3 - Detection and Tracking

Now that we have our BEV video, we can use our yolov8 model to detect any fish in the video and associate them into tracks with DeepSORT. Make sure that the detection model path, confidence threshold, and visualization parameters are set.



```
# input video directory
input_dir: "F:/ChuteFishFullPipeline/demo/test_video"
# output directory
output_dir: "F:/ChuteFishFullPipeline/demo/test_out"
# path to camera calibration file
calib_path: "F:/ChuteFishFullPipeline/demo/calibration/cameraParamBEV-alpha-0.2.xml"

det_model_path: "yolov8_chute.pt"

save_vid: True
fps: [illegible]

det_thresh: [illegible]

class_path: chute2022.pth

segment_model: "h"
# Whether to perform segmentation or just visualize detection and classification results
segment: True
# number of points for midline
midline_pts: 10
```

We need to navigate to the detection director in order to run the code without errors.

In [20]:
```
%%cmd
cd ../detection
python track.py --config_path ../demo/demoChuteConfig.yml
```

```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

(yolov8) F:\ChuteFishFullPipeline\demo>cd ../detection

(yolov8) F:\ChuteFishFullPipeline\detection>python track.py --config_path ../demo/demoChuteConfig.yml
```

```
F:\ChuteFishFullPipeline\detection\deep_sort/deep/reid\torchreid\metrics\rank.py:11: UserWarning: Cython evaluation
(very fast so highly recommended) is unavailable, now use python evaluation.
  warnings.warn(
YOLOv5  e1c5877 torch 1.10.1 CUDA:0 (NVIDIA GeForce RTX 2070 SUPER, 8192MiB)

YOLOv5  e1c5877 torch 1.10.1 CUDA:0 (NVIDIA GeForce RTX 2070 SUPER, 8192MiB)
```

```
../demo/demoChuteConfig.yml
0.5

0: 1088x1088 (no detections), 22.0ms
Speed: 0.0ms preprocess, 22.0ms inference, 3.0ms postprocess per image at shape (1, 3, 1088, 1088)
```

```
video 1/1 (1/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.025s),
Tracking:(0.025s)
video 1/1 (2/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.017s),
Tracking:(0.029s)
video 1/1 (3/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.016s),
Tracking:(0.026s)
video 1/1 (4/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.028s)
video 1/1 (5/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.018s),
Tracking:(0.024s)
video 1/1 (6/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.017s),
Tracking:(0.024s)
video 1/1 (7/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.017s),
Tracking:(0.023s)
video 1/1 (8/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.018s),
Tracking:(0.030s)
video 1/1 (9/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.018s),
Tracking:(0.025s)
video 1/1 (10/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.017s),
Tracking:(0.021s)
video 1/1 (11/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.022s)
video 1/1 (12/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.018s),
Tracking:(0.023s)
video 1/1 (13/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.021s)
video 1/1 (14/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.018s),
Tracking:(0.022s)
video 1/1 (15/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.022s)
video 1/1 (16/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.023s)
video 1/1 (17/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.022s)
video 1/1 (18/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.022s)
video 1/1 (19/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.017s),
Tracking:(0.023s)
video 1/1 (20/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.016s),
Tracking:(0.024s)
video 1/1 (21/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 1 fish, Done. YOLO:(0.017s),
Tracking:(0.026s)
video 1/1 (22/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.016
s), Tracking:(0.024s)
video 1/1 (23/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.018
s), Tracking:(0.025s)
video 1/1 (24/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.017
s), Tracking:(0.025s)
video 1/1 (25/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.016
s), Tracking:(0.025s)
video 1/1 (26/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 2 fishs, Done. YOLO:(0.016
s), Tracking:(0.025s)
video 1/1 (27/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.015
s), Tracking:(0.030s)
video 1/1 (28/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.015
s), Tracking:(0.027s)
video 1/1 (29/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.016
s), Tracking:(0.028s)
video 1/1 (30/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.016
s), Tracking:(0.028s)
video 1/1 (31/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.021
s), Tracking:(0.026s)
video 1/1 (32/32) F:\ChuteFishFullPipeline\demo\test_out\undistorted\test2.mp4: 480x1088 3 fishs, Done. YOLO:(0.074
s), Tracking:(0.031s)
Speed: 0.847ms pre-process, 18.782ms inference, 0.031ms NMS, 24.982ms deep sort update, 58.353ms post-process
per image at shape (1, 3, 1088, 1088)
```

```
Successfully loaded imagenet pretrained weights from "C:\Users\aotia/.cache\torch\checkpoints\osnet_x0_25_imagenet.pt
h"
** The following layers are discarded due to unmatched keys or layer size: ['classifier.weight', 'classifier.bias']
Model: osnet_x0_25
- params: 203,568
- flops: 82,316,000
Results saved to F:\ChuteFishFullPipeline\demo\test_out\detection\test2.mp4
```

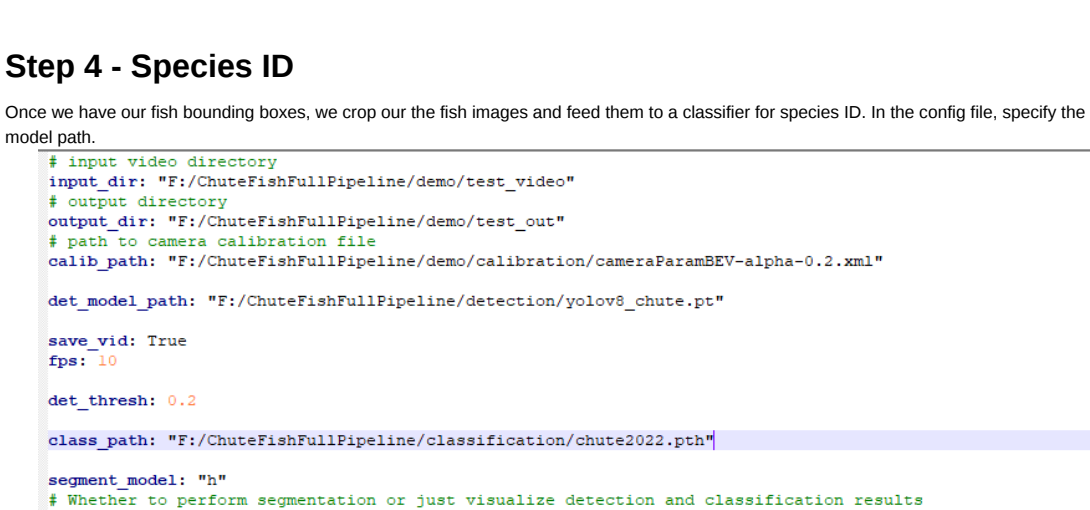
```
OpenCV: FFMPEG: tag 0x30395056/'VP90' is not supported with codec id 167 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x39307076/'vp09'
```

```
(yolov8) F:\ChuteFishFullPipeline\detection>
```

In [ ]:

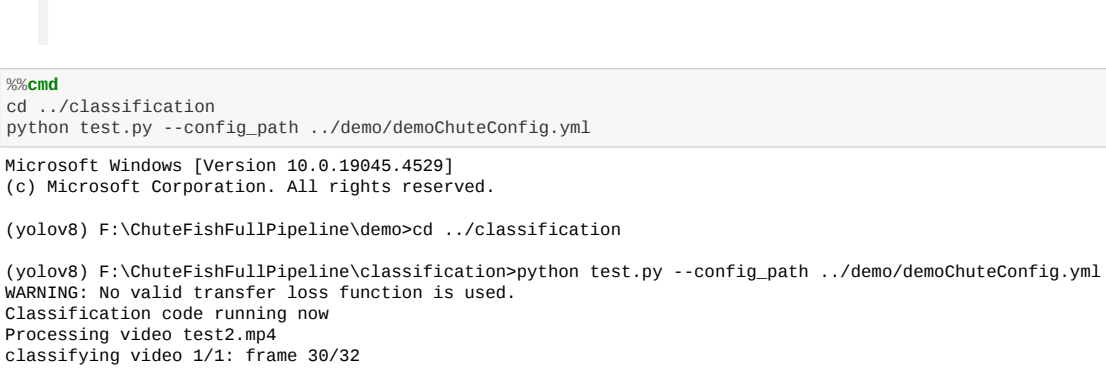
The detection and tracking results should be saved to F:\ChuteFishFullPipeline\demo\test_out\detection

In [4]:
```python
from ipywidgets import Video
Video.from_file("./test_out/detection/test2.mp4", width=640)
```



Step 4 - Species ID

Once we have our fish bounding boxes, we crop our the fish images and feed them to a classifier for species ID. In the config file, specify the classification model path.

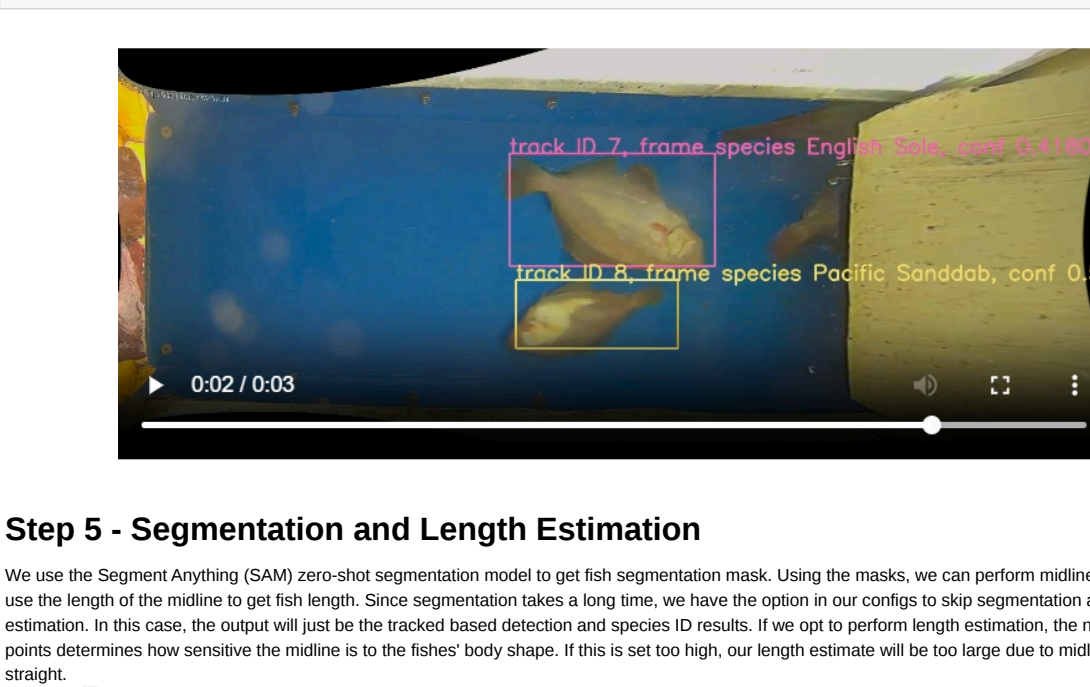


```
# input video directory
input_dir: "F:/ChuteFishFullPipeline/demo/test_video"
# output directory
output_dir: "F:/ChuteFishFullPipeline/demo/test_out"
# path to camera calibration file
calib_path: "F:/ChuteFishFullPipeline/demo/calibration/cameraParamBEV-alpha-0.2.xml"

det_model_path: "F:/ChuteFishFullPipeline/detection/yolov8_chute.pt"

save_vid: True
fps: 10

det_thresh: 0.2

class_path: "F:/ChuteFishFullPipeline/classification/chute2022.pth"

segment_model: "h"
# Whether to perform segmentation or just visualize detection and classification results
segment: True
# number of points for midline
midline_pts: 10
```

In [22]:
```
%%cmd
cd ../classification
python test.py --config_path ../demo/demoChuteConfig.yml
```

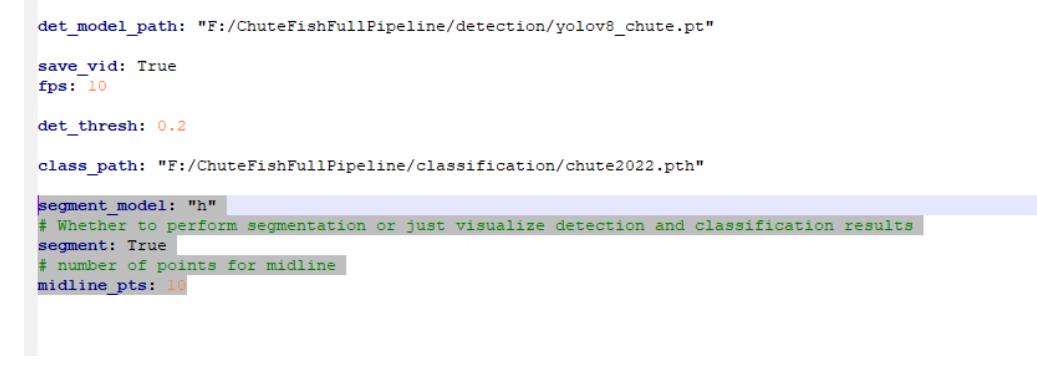
```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

(yolov8) F:\ChuteFishFullPipeline\demo>cd ../classification

(yolov8) F:\ChuteFishFullPipeline\classification>python test.py --config_path ../demo/demoChuteConfig.yml
WARNING: No valid transfer loss function is used.
Classification code running now
Processing video test2.mp4
classifying video 1/1: frame 30/32
(yolov8) F:\ChuteFishFullPipeline\classification>
```

In [5]:
```python
from ipywidgets import Video
Video.from_file("./test_out/classification/test2.mp4", width=640)
```


track ID 7, frame species English Sole, conf
track ID 8, frame species Pacific Sanddab, conf 0.58
0:02 / 0:03

Step 5 - Segmentation and Length Estimation

We use the Segment Anything (SAM) zero-shot segmentation model to get fish segmentation mask. Using the masks, we can perform midline estimation and use the length of the midline to get fish length. Since segmentation takes a long time, we have the option in our configs to skip segmentation and length estimation. In this case, the output will just be the tracked based detection and species ID results. If we opt to perform length estimation, the number of midline points determines how sensitive the midline is to the fishes' body shape. If this is set too high, our length estimate will be too large due to midline not being straight.



```
# input video directory
input_dir: "F:/ChuteFishFullPipeline/demo/test_video"
# output directory
output_dir: "F:/ChuteFishFullPipeline/demo/test_out"
# path to camera calibration file
calib_path: "F:/ChuteFishFullPipeline/demo/calibration/cameraParamBEV-alpha-0.2.xml"

det_model_path: "F:/ChuteFishFullPipeline/detection/yolov8_chute.pt"

save_vid: True
fps: 10

det_thresh: 0.2

class_path: "F:/ChuteFishFullPipeline/classification/chute2022.pth"

segment_model: "h"
# Whether to perform segmentation or just visualize detection and classification results
segment: True
# number of points for midline
midline_pts: [illegible]
```

You can download the SAM models from https://github.com/facebookresearch/segment-anything

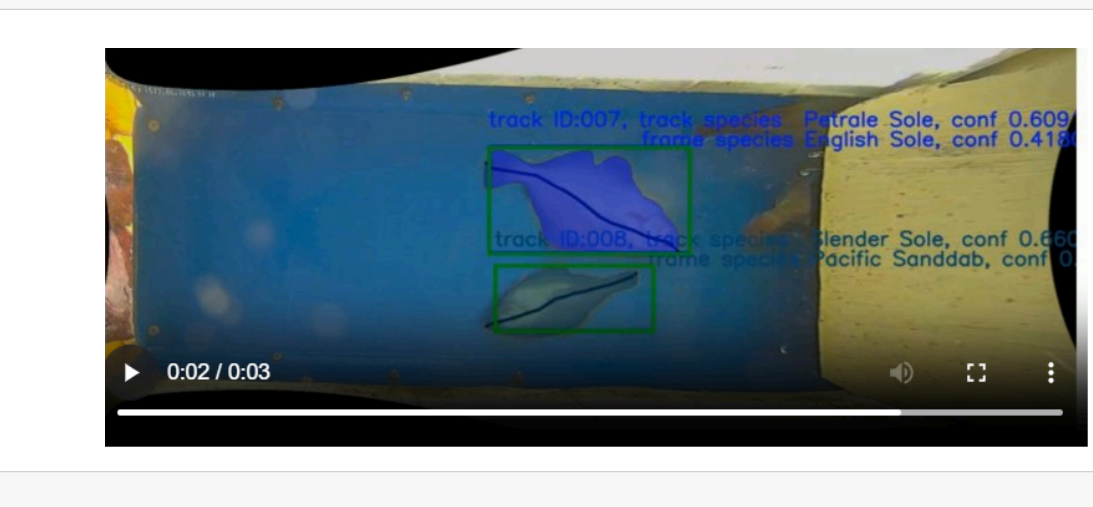
In [24]:
```
%%cmd
cd ../segmentation
python segment.py --config_path ../demo/demoChuteConfig.yml
```

```
Microsoft Windows [Version 10.0.19045.4529]
(c) Microsoft Corporation. All rights reserved.

(yolov8) F:\ChuteFishFullPipeline\demo>cd ../segmentation

(yolov8) F:\ChuteFishFullPipeline\segmentation>python segment.py --config_path ../demo/demoChuteConfig.yml
Segmenting video test2.mp4
```

```
OpenCV: FFMPEG: tag 0x30395056/'VP90' is not supported with codec id 167 and format 'mp4 / MP4 (MPEG-4 Part 14)'
OpenCV: FFMPEG: fallback to use tag 0x39307076/'vp09'
```

```
Segmenting video 1/1: Frame 33, avg seconds per frame: 1.955
(yolov8) F:\ChuteFishFullPipeline\segmentation>
```

In [6]:
```python
from ipywidgets import Video
Video.from_file("./test_out/segmentation/test2.mp4", width=640)
```



In [ ]:
~~~